

Requirements Specification

Final Updated Requirements for RoboMamba

Oria Cohen | Computer Science Final Project

Project	Multimodal Behavioral Cloning for a Robotic Arm Using Mamba and OpenCLIP
Student	Oria Cohen
Document Type	Final updated requirements specification
System Type	Offline robotic imitation learning pipeline
Primary Model Task	Continuous next-action prediction

1. Purpose

This document defines the final updated requirements for the RoboMamba project. It aligns the original project idea with the implemented system: an offline multimodal robotic imitation learning pipeline that combines YOLO-based data curation, OpenCLIP visual embeddings, robot-state fusion, Mamba sequence modeling, training, evaluation, and Streamlit visualization.

The final system focuses on behavioral cloning for next-action prediction. Success/failure information is used mainly as part of dataset filtering and curation rather than as the main output of the Mamba model.

2. Project Scope

The system processes recorded robotic manipulation trajectories and prepares them for imitation learning. It does not send commands to a physical robot and is not presented as a real-time closed-loop control system.

- Input scope: recorded trajectory folders with image frames and robot-state/action data.
- Processing scope: dataset validation, YOLO filtering, filtered CSV generation, OpenCLIP embedding extraction, and multimodal sequence construction.
- Learning scope: Mamba-based behavioral cloning for continuous 7-dimensional action prediction.
- Evaluation scope: quantitative prediction errors and qualitative prediction examples.
- User interface scope: Streamlit pipeline control, progress display, logs, checkpoint selection, evaluation, and results visualization.

3. Stakeholders

Stakeholder	Role / Interest
Project developer	Implements, runs, validates, and demonstrates the system.
Academic advisor	Reviews the project direction, implementation, documentation, and results.
Academic course framework	Evaluates the final project deliverables and engineering process.
Future students / researchers	May use the repository as a reference for multimodal robotic imitation learning pipelines.

4. Assumptions

- The system operates on offline recorded robotic trajectories rather than live robot control data.
- Each trajectory includes image frames and robot-state/action information that can be aligned by timestep or index.
- OpenCLIP is used as a pretrained visual encoder and remains frozen during the main pipeline.
- YOLO weights adapted to the project environment are available for the data-curation stage.
- The development environment includes Python, PyTorch, and preferably a CUDA-capable GPU.
- The final prediction task is continuous next-action prediction using behavioral cloning.

5. Functional Requirements

FR-01 - Dataset Selection and Validation | Priority: High | Verification: Test

The system shall allow the user to select a dataset folder and shall validate that the selected folder contains a usable trajectory structure. If the selected folder is invalid or incomplete, the system shall prevent progress to dependent stages.

FR-02 - Existing CSV Detection and Reuse | Priority: High | Verification: Test

The system shall detect an existing filtered CSV file, validate required columns, display its status, and allow reuse when the CSV is valid.

FR-03 - YOLO-Based Data Curation | Priority: High | Verification: Test

The system shall integrate YOLO-based filtering to support visual validation of trajectories using gripper and object detections. This component shall support the project-specific detector weights produced through manual annotation and training/fine-tuning.

FR-04 - Conservative Filtering Policy | Priority: High | Verification: Inspection / Test

The system shall apply filtering conditions that prefer rejecting ambiguous trajectories over accepting visually invalid or failed trajectories as successful examples.

FR-05 - Filtered CSV Generation | Priority: High | Verification: Test

The system shall generate a filtered CSV containing trajectory paths and status information for downstream OpenCLIP, training, and evaluation stages.

FR-06 - OpenCLIP Embedding Generation | Priority: High | Verification: Test

The system shall encode image frames into fixed-size visual embeddings using a pretrained OpenCLIP model.

FR-07 - Embedding Cache Management | Priority: High | Verification: Test

The system shall save embeddings to cache, detect existing caches, support reuse, and allow recomputation when requested.

FR-08 - Multimodal Fusion | Priority: High | Verification: Test

The system shall concatenate 512-dimensional visual embeddings with 7-dimensional robot-state vectors to create 519-dimensional timestep representations.

FR-09 - Sequence Window Construction | Priority: High | Verification: Test

The system shall build fixed-length temporal windows from filtered trajectories for training and evaluation.

FR-10 - Mamba-Based Action Prediction | Priority: High | Verification: Test

The system shall train or load a Mamba-based model that predicts the next continuous 7-dimensional robot action.

FR-11 - Training Workflow | Priority: High | Verification: Test

The system shall support hyperparameter configuration, train/validation splitting, loss calculation, early stopping, checkpoint saving, metrics saving, and logging.

FR-12 - Checkpoint Selection | Priority: High | Verification: Test

The system shall detect available checkpoints, display basic metadata, allow selecting a checkpoint, and prevent evaluation when no valid checkpoint is selected.

FR-13 - Evaluation Metrics | Priority: High | Verification: Test

The system shall compute evaluation metrics such as MAE, MSE, per-dimension MAE, and sample counts.

FR-14 - Prediction Examples | Priority: Medium | Verification: Test

The system shall save and display prediction examples comparing predicted actions with ground-truth actions.

FR-15 - Results Visualization | Priority: High | Verification: Test

The system shall display metrics, charts, action comparison tables, trajectory images, prediction examples, and evaluation logs.

FR-16 - Streamlit UI Flow | Priority: High | Verification: Inspection / Test

The system shall provide a staged Streamlit interface for Dataset, YOLO, OpenCLIP, Model/Training, Evaluation, and Results.

FR-17 - Logging and Error Handling | Priority: High | Verification: Test

The system shall display clear logs and user-facing error messages for missing files, invalid inputs, failed stages, and incomplete artifacts.

6. Non-Functional Requirements

ID	Quality Attribute	Description	Priority	Verification
NFR-01	Performance	The pipeline should use GPU acceleration when available for OpenCLIP, Mamba, and training workloads.	High	Test / Report
NFR-02	Reproducibility	Training and evaluation runs should save configuration, checkpoint, metrics, and log artifacts.	High	Inspection
NFR-03	Modularity	The system should keep dataset loading, YOLO filtering, OpenCLIP extraction, Mamba modeling, training, evaluation, and UI logic separated into modules.	High	Inspection
NFR-04	Maintainability	The code should be organized in a clear repository structure with reusable modules and documented execution paths.	Medium	Inspection
NFR-05	Robustness	The system should handle missing datasets, invalid CSV files, missing embeddings, and missing checkpoints without misleading the user.	High	Test
NFR-06	Observability	The system should provide logs, progress information, metrics, status indicators, and result artifacts throughout the pipeline.	Medium	Inspection / Test
NFR-07	Data Handling	Large datasets and generated artifacts should remain local and should not be committed to Git by default.	Medium	Inspection
NFR-08	Usability	The UI should present the pipeline stages, results, and errors clearly enough for final demonstration and review.	Medium	Inspection / Test

7. Technology Requirements

Category	Requirement
Programming Language	Python 3.10+
Deep Learning Framework	PyTorch with CUDA support when available
Visual Encoder	OpenCLIP ViT-B-32 pretrained model for image embeddings
Object Detection / Curation	YOLO model adapted to the project dataset, including gripper and object detections
Sequence Model	Mamba / Mamba-SSM based model for temporal sequence processing
Data Processing	NumPy, Pandas, PIL/OpenCV-style image processing utilities

Category	Requirement
User Interface	Streamlit for staged workflow execution and visualization
Visualization	Plotly and tabular result views
Experiment Artifacts	Local files for CSVs, embeddings, checkpoints, configs, logs, metrics, and prediction examples
Version Control	GitHub repository with source code and final documentation; large generated artifacts excluded by default

8. Dataset and Validation Requirements

The system is designed around a selected large BridgeData-style subset of recorded robotic manipulation trajectories. The dataset used during development contained approximately 8,000 trajectory folders before filtering and approximately 4,100 successful trajectories after applying the filtering logic.

YOLO filtering results were manually inspected on a representative sample that included approximately 80 folders classified as successful and approximately 80 folders classified as failed. This inspection helped validate that the filtering decisions were reasonable and that the downstream dataset was suitable for behavioral cloning training.

9. Items Outside the Final Implementation Scope

Item	Clarification
Physical Robot Deployment	The system does not deploy the trained model on a physical robotic arm.
Real-Time Closed-Loop Control	The system does not send real-time commands to a robot; it evaluates predictions offline on recorded trajectories.
Success/Failure as Main Mamba Output	Success/failure information is used mainly for filtering and curation; the final Mamba task is continuous action prediction.
OpenCLIP Fine-Tuning	OpenCLIP remains frozen; the main learning component is the Mamba sequence model.
Full Baseline Benchmarking	The final implementation does not include a full experimental benchmark against RNN, LSTM, or Transformer baselines.
Full BridgeData Benchmark	The project focuses on the selected dataset subset rather than evaluating all BridgeData tasks and variants.
Authentication and User Permissions	The tool is intended for local research use and does not include user-management or security mechanisms.

10. Traceability to Final Implementation

Area	Implemented Coverage
Data preparation	Dataset validation, YOLO filtering, filtered CSV creation, embedding cache detection and reuse.
Visual learning component	OpenCLIP extracts stable visual embeddings from image frames.
Multimodal representation	Visual embeddings are concatenated with robot-state vectors to form 519-dimensional inputs.
Temporal learning component	Mamba processes fixed-length temporal windows and predicts the next robot action.
Evaluation	MAE, MSE, per-dimension MAE, sample counts, and prediction examples are produced.
User workflow	Streamlit UI controls the staged pipeline and displays logs, progress, checkpoints, results, charts, and examples.

11. Summary

These requirements define the final updated scope of the RoboMamba system. The project is centered on a complete offline multimodal behavioral cloning pipeline: data curation with YOLO, visual embedding with OpenCLIP, robot-state fusion, sequence learning with Mamba, evaluation, and Streamlit-based result presentation.